

- 7 Nuclei of an isotope of copper (Cu) each have 29 protons and 37 neutrons. This isotope is a β^- emitter.

(a) State the nuclide notation in the form ${}^A_Z\text{X}$ for this nucleus of copper.

[1]

- (b) The energy spectrum of the β^- radiation emitted by a sample of this isotope is shown in Fig. 7.1.

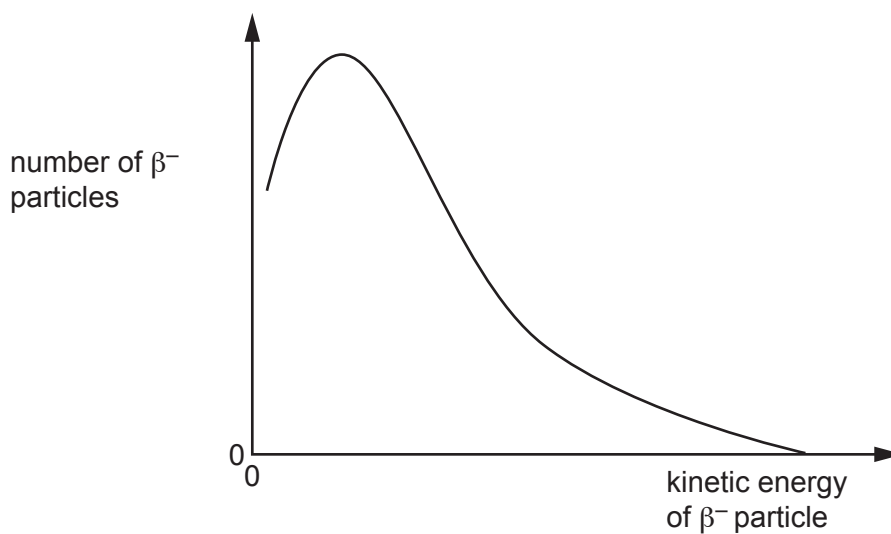


Fig. 7.1

- (i) Use Fig. 7.1 to explain why other particles apart from the β^- particles must be emitted during this decay.

.....

 [3]

- (ii) State the name of the other particle emitted during the decay of this isotope.

..... [1]

- (iii) The copper isotope decays to an isotope of zinc (Zn).

Give the radioactive decay equation for this decay. Include the nucleon and proton numbers of **all** the particles involved.

[3]

[Total: 8]